

# CLINICAL STUDY SERIES

## DISEASE CONCEPT MAPS (PRE-FILLED)

1 Disease Per Page · NCLEX Essentials · 36 Printable Pages

### HOW TO USE

- Print at 100% / "Actual Size" on US Letter paper.
- Work from memory first, then check your notes and textbook.
- Reprint blank copies for each drug class or exam unit.

# Hypertension

**WHAT IS IT?**

Persistently raised blood pressure (130/80 mmHg or higher on repeated readings).

**PATHOPHYSIOLOGY**

Increased vascular resistance and/or volume raises arterial pressure, damaging vessel walls and straining the heart over time.

**CAUSES & RISK FACTORS**

Primary: age, genetics, obesity, high sodium intake, inactivity, alcohol, smoking. Secondary: kidney disease, endocrine disorders, sleep apnea.

**SIGNS & SYMPTOMS**

Usually no symptoms ('silent killer'). Severe: headache, blurred vision, chest pain, nosebleeds.

**DIAGNOSTICS & LABS**

Repeated BP readings (office, home or ambulatory). BMP, lipids, glucose, urinalysis, ECG to look for organ damage.

**TREATMENT & MEDICATIONS**

Lifestyle changes (DASH diet, less sodium, exercise). Thiazides, ACE inhibitors/ARBs, calcium channel blockers; add others as needed.

**NURSING CARE**

Correct cuff size, rested seated readings. Watch for orthostatic hypotension with new meds. Assess adherence and side effects.

**COMPLICATIONS**

Stroke, MI, heart failure, kidney disease, retinopathy, aortic aneurysm.

**PATIENT EDUCATION**

Take meds daily even when feeling well. Home BP log, limit sodium and alcohol, stop smoking, rise slowly.

# Heart Failure

**WHAT IS IT?**

The heart cannot pump enough blood to meet the body's needs (reduced or preserved ejection fraction).

**PATHOPHYSIOLOGY**

Weak or stiff ventricle lowers output; RAAS and sympathetic activation retain fluid and increase workload, worsening failure.

**CAUSES & RISK FACTORS**

Coronary artery disease, prior MI, hypertension, valve disease, cardiomyopathy, arrhythmias.

**SIGNS & SYMPTOMS**

Left: dyspnea, orthopnea, crackles, fatigue. Right: edema, JVD, weight gain, ascites, hepatomegaly.

**DIAGNOSTICS & LABS**

BNP/NT-proBNP, echocardiogram (ejection fraction), chest X-ray, ECG, BMP.

**TREATMENT & MEDICATIONS**

Diuretics, ACE inhibitor/ARB/ARNI, beta blocker, aldosterone antagonist, SGLT2 inhibitor. Sodium and fluid limits.

**NURSING CARE**

Daily weight same time and scale, strict I&O, lung sounds, edema checks, monitor K+ and kidney function.

**COMPLICATIONS**

Pulmonary edema, arrhythmias, kidney injury, cardiogenic shock.

**PATIENT EDUCATION**

Report gain of 2-3 lb in a day or 5 lb in a week. Low sodium diet, take meds as prescribed, pace activities.

# Myocardial Infarction

**WHAT IS IT?**

Death of heart muscle from blocked coronary blood flow (heart attack).

**PATHOPHYSIOLOGY**

Plaque ruptures, a clot forms and blocks a coronary artery; the muscle it feeds becomes ischemic and dies without reperfusion.

**CAUSES & RISK FACTORS**

Atherosclerosis. Risks: smoking, hypertension, diabetes, high LDL, obesity, family history, age, male sex.

**SIGNS & SYMPTOMS**

Chest pressure radiating to arm, jaw or back; dyspnea, sweating, nausea. Women, older adults and diabetics may have atypical symptoms.

**DIAGNOSTICS & LABS**

12-lead ECG within 10 minutes (ST elevation = STEMI), serial troponins, echocardiogram, cardiac catheterization.

**TREATMENT & MEDICATIONS**

Aspirin, nitroglycerin, oxygen if low SpO<sub>2</sub>, anticoagulant, P2Y<sub>12</sub> inhibitor. STEMI: PCI or thrombolytics. Then beta blocker, statin, ACE inhibitor.

**NURSING CARE**

Continuous ECG, vital signs, pain assessment, IV access, time-critical care. Monitor for arrhythmias and bleeding.

**COMPLICATIONS**

Arrhythmias, heart failure, cardiogenic shock, papillary muscle rupture, pericarditis.

**PATIENT EDUCATION**

Cardiac rehab, take antiplatelets without stopping, heart-healthy diet, call 911 for chest pain not relieved in 5 minutes.

# Atrial Fibrillation

**WHAT IS IT?**

Irregular, rapid atrial rhythm with no effective atrial contraction.

**PATHOPHYSIOLOGY**

Chaotic atrial electrical activity; blood pools in the atria (especially the left atrial appendage), allowing clots to form.

**CAUSES & RISK FACTORS**

Age, hypertension, heart failure, valve disease, hyperthyroidism, alcohol, sleep apnea, surgery.

**SIGNS & SYMPTOMS**

Irregularly irregular pulse, palpitations, fatigue, dizziness, dyspnea; may have no symptoms.

**DIAGNOSTICS & LABS**

ECG: no P waves, irregular R-R intervals. Echocardiogram, thyroid tests, CHA2DS2-VASc stroke risk score.

**TREATMENT & MEDICATIONS**

Rate control (beta blocker, diltiazem, digoxin), rhythm control (amiodarone, cardioversion, ablation), anticoagulation (DOAC or warfarin).

**NURSING CARE**

Apical pulse for a full minute, monitor for stroke signs, bleeding precautions on anticoagulants.

**COMPLICATIONS**

Stroke, heart failure, clots elsewhere, bleeding from anticoagulants.

**PATIENT EDUCATION**

Take anticoagulant exactly as prescribed, know stroke signs (BE FAST), report bleeding, limit alcohol.

# Deep Vein Thrombosis

**WHAT IS IT?**

A blood clot in a deep vein, usually in the leg.

**PATHOPHYSIOLOGY**

Virchow's triad: slow blood flow, vessel wall injury and hypercoagulability lead to clot formation.

**CAUSES & RISK FACTORS**

Immobility, surgery (especially orthopedic), cancer, pregnancy, estrogen therapy, prior clot, obesity.

**SIGNS & SYMPTOMS**

One-sided leg swelling, pain, warmth, redness. Can be silent.

**DIAGNOSTICS & LABS**

Compression (duplex) ultrasound, D-dimer (helps rule out), Wells score.

**TREATMENT & MEDICATIONS**

Anticoagulation (DOAC, LMWH, heparin, warfarin). Thrombolysis or IVC filter in select cases.

**NURSING CARE**

Never massage the leg. Monitor for PE signs, bleeding, platelet count on heparin. Early mobility once anticoagulated.

**COMPLICATIONS**

Pulmonary embolism, post-thrombotic syndrome, recurrence.

**PATIENT EDUCATION**

Take anticoagulants as ordered, move and stretch on long trips, stay hydrated, report sudden breathlessness.

# Pulmonary Embolism

**WHAT IS IT?**

A clot that travels to and blocks a pulmonary artery.

**PATHOPHYSIOLOGY**

A clot (usually from a leg DVT) lodges in the lungs, blocking blood flow and oxygen exchange and straining the right heart.

**CAUSES & RISK FACTORS**

DVT risk factors: immobility, surgery, cancer, pregnancy, estrogen, prior clot.

**SIGNS & SYMPTOMS**

Sudden dyspnea, pleuritic chest pain, tachycardia, tachypnea, anxiety, low SpO<sub>2</sub>, hemoptysis.

**DIAGNOSTICS & LABS**

CT pulmonary angiography, D-dimer (low-risk patients), ABG, ECG, echocardiogram.

**TREATMENT & MEDICATIONS**

Oxygen, anticoagulation; thrombolytics or embolectomy for massive PE with shock.

**NURSING CARE**

High Fowler's, oxygen, continuous monitoring, IV access, prepare for emergency treatment, bleeding precautions.

**COMPLICATIONS**

Right heart failure, shock, death, chronic pulmonary hypertension.

**PATIENT EDUCATION**

Continue anticoagulants, prevent DVT (movement, hydration, compression if ordered), know warning signs.

# Ischemic Stroke

**WHAT IS IT?**

Loss of brain function from a blocked artery supplying the brain.

**PATHOPHYSIOLOGY**

A clot or narrowing stops blood flow; brain cells die within minutes, with a salvageable penumbra around the core.

**CAUSES & RISK FACTORS**

Atrial fibrillation (embolic), atherosclerosis, hypertension, diabetes, smoking, high cholesterol.

**SIGNS & SYMPTOMS**

BE FAST: Balance, Eyes, Face droop, Arm weakness, Speech difficulty, Time to call 911.

**DIAGNOSTICS & LABS**

Non-contrast CT first (rule out bleed), CT angiography, MRI, glucose check, NIH Stroke Scale.

**TREATMENT & MEDICATIONS**

Alteplase/tenecteplase within the time window, thrombectomy for large vessel occlusion, then antiplatelets, statin, BP control.

**NURSING CARE**

Know last known well time, neuro checks, swallow screen before any oral intake, BP limits per protocol, fall precautions.

**COMPLICATIONS**

Aspiration pneumonia, brain swelling, hemorrhagic conversion, DVT, depression.

**PATIENT EDUCATION**

BE FAST warning signs, control BP and diabetes, take antiplatelets or anticoagulants, rehab participation.

# Type 2 Diabetes Mellitus

**WHAT IS IT?**

Chronic high blood glucose from insulin resistance and declining insulin production.

**PATHOPHYSIOLOGY**

Cells respond poorly to insulin; the pancreas cannot keep up, so glucose builds up and damages blood vessels and nerves.

**CAUSES & RISK FACTORS**

Obesity, inactivity, family history, age, certain ethnic backgrounds, gestational diabetes history.

**SIGNS & SYMPTOMS**

Polyuria, polydipsia, fatigue, blurred vision, slow-healing wounds, frequent infections; often no symptoms early.

**DIAGNOSTICS & LABS**

A1C 6.5% or higher, fasting glucose 126 mg/dL or higher, 2-hr OGTT 200 or higher, random 200+ with symptoms.

**TREATMENT & MEDICATIONS**

Diet and exercise, metformin, then GLP-1 agonists, SGLT2 inhibitors, DPP-4 inhibitors, sulfonylureas or insulin.

**NURSING CARE**

Glucose monitoring, hypoglycemia recognition, foot checks, insulin technique, A1C goals.

**COMPLICATIONS**

Heart disease, stroke, kidney disease, retinopathy, neuropathy, foot ulcers, HHS.

**PATIENT EDUCATION**

Glucose self-checks, rule of 15 for lows, sick-day rules, daily foot care, regular eye exams.

# Diabetic Ketoacidosis (DKA)

**WHAT IS IT?**

Life-threatening high glucose with ketone production and metabolic acidosis.

**PATHOPHYSIOLOGY**

Severe insulin lack forces fat breakdown into ketones, causing acidosis; high glucose causes osmotic diuresis and dehydration.

**CAUSES & RISK FACTORS**

Type 1 diabetes, missed insulin, infection, new diagnosis, illness or stress. Can occur with SGLT2 inhibitors.

**SIGNS & SYMPTOMS**

Polyuria, thirst, nausea/vomiting, abdominal pain, fruity breath, Kussmaul breathing, confusion.

**DIAGNOSTICS & LABS**

Glucose usually over 250, pH under 7.3, bicarbonate under 18, ketones in blood/urine, raised anion gap.

**TREATMENT & MEDICATIONS**

IV fluids first, IV regular insulin infusion, potassium replacement, add dextrose as glucose falls.

**NURSING CARE**

Hourly glucose, check K<sup>+</sup> before insulin (hold if under 3.3), strict I&O, neuro checks, cardiac monitoring.

**COMPLICATIONS**

Hypokalemia, hypoglycemia, cerebral edema (especially children), shock.

**PATIENT EDUCATION**

Never skip insulin when sick, check ketones when glucose high, sick-day plan, seek help early.

# Hypothyroidism

**WHAT IS IT?**

Too little thyroid hormone, slowing body metabolism.

**PATHOPHYSIOLOGY**

Low T3/T4 reduces metabolic rate in all tissues; primary disease causes high TSH as the pituitary tries to compensate.

**CAUSES & RISK FACTORS**

Hashimoto's thyroiditis (most common), thyroid surgery or radiation, iodine deficiency, lithium, amiodarone.

**SIGNS & SYMPTOMS**

Fatigue, weight gain, cold intolerance, constipation, dry skin, hair loss, bradycardia, depression.

**DIAGNOSTICS & LABS**

High TSH with low free T4 (primary). Thyroid antibodies for Hashimoto's.

**TREATMENT & MEDICATIONS**

Levothyroxine, adjusted every 6-8 weeks by TSH.

**NURSING CARE**

Start low in older adults and heart disease (angina risk). Watch for signs of excess dosing. Monitor TSH.

**COMPLICATIONS**

Myxedema coma (hypothermia, bradycardia, altered mental status), heart disease, infertility.

**PATIENT EDUCATION**

Take levothyroxine on an empty stomach at the same time daily, lifelong therapy, report palpitations.

# Hyperthyroidism (Graves' Disease)

**WHAT IS IT?**

Too much thyroid hormone, speeding up body metabolism.

**PATHOPHYSIOLOGY**

Excess T3/T4 increases metabolic rate and sympathetic effects; in Graves' antibodies stimulate the TSH receptor.

**CAUSES & RISK FACTORS**

Graves' disease (most common), toxic nodules, thyroiditis, excess thyroid medication.

**SIGNS & SYMPTOMS**

Weight loss, heat intolerance, tachycardia, tremor, anxiety, diarrhea, goiter; exophthalmos in Graves'.

**DIAGNOSTICS & LABS**

Low TSH with high free T4/T3, TSH receptor antibodies, radioactive iodine uptake scan.

**TREATMENT & MEDICATIONS**

Methimazole (PTU in first trimester), beta blockers for symptoms, radioactive iodine or thyroidectomy.

**NURSING CARE**

Monitor HR and weight, cool calm environment, high-calorie diet, eye protection, watch for thyroid storm.

**COMPLICATIONS**

Thyroid storm (fever, tachycardia, delirium), AFib, osteoporosis, vision problems.

**PATIENT EDUCATION**

Report fever or sore throat on antithyroid drugs (agranulocytosis), take meds consistently, follow-up labs.

**WHAT IS IT?**

Chronic airway inflammation with reversible bronchospasm.

**PATHOPHYSIOLOGY**

Triggers cause airway inflammation, bronchoconstriction and mucus production, narrowing airflow.

**CAUSES & RISK FACTORS**

Allergens, respiratory infections, exercise, cold air, smoke, stress, NSAIDs/aspirin in some people.

**SIGNS & SYMPTOMS**

Wheezing, cough (often at night), chest tightness, dyspnea. Severe: silent chest, inability to speak.

**DIAGNOSTICS & LABS**

Spirometry with bronchodilator reversibility, peak flow, allergy testing.

**TREATMENT & MEDICATIONS**

Reliever (SABA or ICS-formoterol), controller inhaled corticosteroid, add LABA, leukotriene antagonist, biologics.

**NURSING CARE**

Assess breath sounds and SpO<sub>2</sub>, high Fowler's, give bronchodilators, watch for fatigue and a silent chest.

**COMPLICATIONS**

Status asthmaticus, respiratory failure, pneumothorax.

**PATIENT EDUCATION**

Inhaler and spacer technique, asthma action plan, avoid triggers, rinse mouth after steroid inhalers.

**WHAT IS IT?**

Progressive, mostly irreversible airflow limitation (emphysema and chronic bronchitis).

**PATHOPHYSIOLOGY**

Chronic irritation destroys alveoli and inflames airways, trapping air and reducing gas exchange.

**CAUSES & RISK FACTORS**

Smoking (main cause), occupational dusts and chemicals, air pollution, alpha-1 antitrypsin deficiency.

**SIGNS & SYMPTOMS**

Chronic cough, sputum, dyspnea on exertion, barrel chest, pursed-lip breathing, prolonged expiration.

**DIAGNOSTICS & LABS**

Spirometry: FEV1/FVC under 0.70 after bronchodilator. Chest X-ray, ABG, SpO2.

**TREATMENT & MEDICATIONS**

Bronchodilators (LAMA, LABA), inhaled steroids for exacerbators, pulmonary rehab, oxygen if hypoxemic, vaccines.

**NURSING CARE**

Target SpO2 88-92% when prescribed, pursed-lip breathing, tripod positioning, nutrition, infection signs.

**COMPLICATIONS**

Exacerbations, respiratory failure, cor pulmonale, pneumonia.

**PATIENT EDUCATION**

Smoking cessation, inhaler technique, energy conservation, vaccinations, recognize exacerbation early.

# Pneumonia

**WHAT IS IT?**

Infection of the lung tissue causing inflammation and fluid in the alveoli.

**PATHOPHYSIOLOGY**

Pathogens reach the alveoli; inflammation fills them with fluid and cells, impairing gas exchange.

**CAUSES & RISK FACTORS**

Bacteria (*Streptococcus pneumoniae* most common), viruses, fungi, aspiration. Risks: age, smoking, immobility, dysphagia.

**SIGNS & SYMPTOMS**

Fever, productive cough, pleuritic chest pain, dyspnea, crackles, tachypnea; older adults may present with confusion.

**DIAGNOSTICS & LABS**

Chest X-ray, CBC, sputum and blood cultures (before antibiotics), SpO<sub>2</sub>/ABG, procalcitonin.

**TREATMENT & MEDICATIONS**

Antibiotics (by setting and risk), oxygen, fluids, fever control.

**NURSING CARE**

Cultures before antibiotics, elevate HOB, cough/deep breathe, incentive spirometer, mobilize, aspiration precautions.

**COMPLICATIONS**

Sepsis, respiratory failure, pleural effusion, empyema.

**PATIENT EDUCATION**

Finish antibiotics, pneumococcal and flu vaccines, stop smoking, rest and hydrate.

# Tuberculosis

**WHAT IS IT?**

Airborne bacterial infection (*Mycobacterium tuberculosis*), mainly of the lungs.

**PATHOPHYSIOLOGY**

Inhaled bacteria form granulomas; they may stay latent or become active disease that damages lung tissue.

**CAUSES & RISK FACTORS**

Close contact with active TB. Risks: HIV, crowded living, incarceration, homelessness, travel to endemic areas.

**SIGNS & SYMPTOMS**

Cough over 3 weeks, hemoptysis, night sweats, fever, weight loss, fatigue.

**DIAGNOSTICS & LABS**

TB skin test or IGRA (infection), chest X-ray, sputum AFB smears and culture x3, NAAT.

**TREATMENT & MEDICATIONS**

Active TB: RIPE therapy (rifampin, isoniazid, pyrazinamide, ethambutol) for 6+ months. Latent: shorter regimens.

**NURSING CARE**

Airborne precautions, negative-pressure room, N95 respirator, directly observed therapy, monitor LFTs.

**COMPLICATIONS**

Drug resistance, spread to other organs, liver toxicity from drugs.

**PATIENT EDUCATION**

Take every dose, cover coughs, orange body fluids with rifampin, avoid alcohol, vitamin B6 with isoniazid.

# Acute Kidney Injury

**WHAT IS IT?**

Sudden loss of kidney function over hours to days.

**PATHOPHYSIOLOGY**

Reduced perfusion (prerenal), kidney tissue damage (intrinsic) or blocked outflow (postrenal) lowers filtration.

**CAUSES & RISK FACTORS**

Dehydration, shock, sepsis, nephrotoxic drugs (NSAIDs, contrast, aminoglycosides, vancomycin), obstruction.

**SIGNS & SYMPTOMS**

Reduced urine output, fluid overload, fatigue, confusion, nausea; may be found only on labs.

**DIAGNOSTICS & LABS**

Rising creatinine, urine output under 0.5 mL/kg/hr, BUN, electrolytes, urinalysis, renal ultrasound.

**TREATMENT & MEDICATIONS**

Treat the cause, fluids if prerenal, stop nephrotoxins, adjust drug doses, dialysis if severe.

**NURSING CARE**

Strict I&O, daily weights, monitor K<sup>+</sup> and ECG, avoid nephrotoxins, watch for fluid overload.

**COMPLICATIONS**

Hyperkalemia, metabolic acidosis, fluid overload, progression to CKD.

**PATIENT EDUCATION**

Stay hydrated, avoid NSAIDs unless approved, follow up on kidney labs.

# Chronic Kidney Disease

**WHAT IS IT?**

Kidney damage or reduced function (GFR under 60) for 3 months or more.

**PATHOPHYSIOLOGY**

Progressive loss of nephrons causes waste retention, fluid and electrolyte imbalance, anemia and bone disease.

**CAUSES & RISK FACTORS**

Diabetes and hypertension (most common), glomerulonephritis, polycystic kidney disease.

**SIGNS & SYMPTOMS**

Early: none. Later: fatigue, edema, itching, nausea, reduced urine, uremic symptoms.

**DIAGNOSTICS & LABS**

eGFR, creatinine, urine albumin-to-creatinine ratio, electrolytes, CBC, renal ultrasound.

**TREATMENT & MEDICATIONS**

Control BP and glucose, ACE inhibitor/ARB, SGLT2 inhibitor, diet limits, ESAs for anemia, phosphate binders, dialysis or transplant.

**NURSING CARE**

Monitor fluid status and electrolytes, protect access arm (no BP or sticks), adjust medication doses.

**COMPLICATIONS**

Hyperkalemia, anemia, bone disease, heart disease, fluid overload.

**PATIENT EDUCATION**

Limit sodium, potassium and phosphorus as advised, avoid NSAIDs, protect fistula, keep dialysis schedule.

**WHAT IS IT?**

Late-stage scarring of the liver that impairs its function.

**PATHOPHYSIOLOGY**

Chronic injury replaces liver cells with scar tissue, raising portal pressure and reducing protein and clotting factor production.

**CAUSES & RISK FACTORS**

Alcohol, hepatitis B and C, fatty liver disease (MASLD), autoimmune and genetic disorders.

**SIGNS & SYMPTOMS**

Fatigue, jaundice, ascites, edema, spider angiomas, easy bruising, confusion, asterixis.

**DIAGNOSTICS & LABS**

LFTs, albumin, INR, bilirubin, platelets, ultrasound, elastography, biopsy.

**TREATMENT & MEDICATIONS**

Treat the cause, no alcohol, diuretics and sodium limit for ascites, lactulose/rifaximin, beta blockers for varices, transplant.

**NURSING CARE**

Monitor mental status, bleeding, weight and abdominal girth, ammonia, avoid sedatives and acetaminophen overuse.

**COMPLICATIONS**

Variceal bleeding, hepatic encephalopathy, SBP, hepatorenal syndrome, liver cancer.

**PATIENT EDUCATION**

No alcohol, low sodium diet, lactulose goal 2-3 soft stools daily, report bleeding or confusion.

# Acute Pancreatitis

**WHAT IS IT?**

Sudden inflammation of the pancreas.

**PATHOPHYSIOLOGY**

Digestive enzymes activate inside the pancreas and digest its own tissue, causing inflammation that can become systemic.

**CAUSES & RISK FACTORS**

Gallstones and alcohol (most common), high triglycerides, some medications, ERCP, trauma.

**SIGNS & SYMPTOMS**

Severe epigastric pain radiating to the back, nausea/vomiting, fever, tachycardia; Cullen's or Grey Turner's sign in severe cases.

**DIAGNOSTICS & LABS**

Lipase over 3 times normal, CT or ultrasound, CBC, calcium, glucose, triglycerides.

**TREATMENT & MEDICATIONS**

IV fluids, pain control, early feeding as tolerated, treat the cause (e.g., cholecystectomy).

**NURSING CARE**

Fluid balance, pain relief, monitor glucose and calcium, positioning (knees to chest), watch for respiratory distress.

**COMPLICATIONS**

Pseudocyst, necrosis, infection, ARDS, shock, diabetes.

**PATIENT EDUCATION**

No alcohol, low-fat diet, report returning pain.

# Peptic Ulcer Disease

**WHAT IS IT?**

Open sores in the stomach or duodenal lining.

**PATHOPHYSIOLOGY**

Acid and pepsin erode mucosa weakened by H. pylori infection or loss of protective prostaglandins.

**CAUSES & RISK FACTORS**

H. pylori infection, NSAIDs, smoking, alcohol, severe physical stress.

**SIGNS & SYMPTOMS**

Burning epigastric pain (gastric: worse with meals; duodenal: relieved by food), bloating, nausea. Bleeding: black stools, coffee-ground vomit.

**DIAGNOSTICS & LABS**

Upper endoscopy, H. pylori breath/stool/biopsy testing, CBC for anemia.

**TREATMENT & MEDICATIONS**

PPIs, H. pylori eradication (antibiotic combination with PPI), stop NSAIDs, endoscopic treatment of bleeding.

**NURSING CARE**

Monitor for GI bleeding and perforation (sudden severe pain, rigid abdomen), hemoglobin, vital signs.

**COMPLICATIONS**

GI bleeding, perforation, gastric outlet obstruction.

**PATIENT EDUCATION**

Finish H. pylori therapy, avoid NSAIDs, stop smoking, limit alcohol, report dark stools.

# Appendicitis

**WHAT IS IT?**

Acute inflammation of the appendix.

**PATHOPHYSIOLOGY**

Blockage of the appendix leads to bacterial overgrowth, swelling and reduced blood flow; it can rupture.

**CAUSES & RISK FACTORS**

Fecalith, lymphoid swelling, infection. Most common in ages 10-30.

**SIGNS & SYMPTOMS**

Pain starting near the navel moving to the right lower quadrant (McBurney's point), anorexia, nausea, fever, rebound tenderness.

**DIAGNOSTICS & LABS**

CBC (raised WBC), CT abdomen (ultrasound in children and pregnancy), pregnancy test in females.

**TREATMENT & MEDICATIONS**

Appendectomy (usually laparoscopic), antibiotics; some uncomplicated cases treated with antibiotics alone.

**NURSING CARE**

NPO, IV fluids, pain control. No heat and no laxatives or enemas. Sudden pain relief may mean rupture.

**COMPLICATIONS**

Perforation, peritonitis, abscess, sepsis.

**PATIENT EDUCATION**

Post-op: early walking, incision care, report fever or increasing pain.

# Inflammatory Bowel Disease

**WHAT IS IT?**

Chronic immune-mediated inflammation of the bowel: Crohn's disease and ulcerative colitis.

**PATHOPHYSIOLOGY**

Crohn's: patchy, full-thickness inflammation anywhere in the GI tract. UC: continuous mucosal inflammation from the rectum up the colon.

**CAUSES & RISK FACTORS**

Genetic, immune and environmental factors; smoking worsens Crohn's.

**SIGNS & SYMPTOMS**

Diarrhea (bloody in UC), abdominal pain, weight loss, fatigue, urgency; Crohn's may cause fistulas.

**DIAGNOSTICS & LABS**

Colonoscopy with biopsy, CBC, CRP, fecal calprotectin, imaging.

**TREATMENT & MEDICATIONS**

Aminosalicylates, corticosteroids for flares, immunomodulators, biologics (anti-TNF, others), surgery.

**NURSING CARE**

Monitor stools, fluids and electrolytes, nutrition, skin around the anus, infection risk on immunosuppressants.

**COMPLICATIONS**

Obstruction, fistulas, toxic megacolon, colon cancer, malnutrition.

**PATIENT EDUCATION**

Recognize flares, follow diet advice, keep vaccinations current, colon cancer screening.

**WHAT IS IT?**

Life-threatening organ dysfunction caused by a dysregulated response to infection.

**PATHOPHYSIOLOGY**

Widespread inflammation causes vasodilation, leaky capillaries and clotting problems, reducing tissue perfusion.

**CAUSES & RISK FACTORS**

Pneumonia, UTI, abdominal and skin infections, lines and catheters. Risks: age, immunosuppression, chronic illness.

**SIGNS & SYMPTOMS**

Fever or hypothermia, tachycardia, tachypnea, hypotension, confusion, reduced urine, mottled skin.

**DIAGNOSTICS & LABS**

Lactate, blood cultures x2 before antibiotics, CBC, BMP, LFTs, coagulation tests, source imaging.

**TREATMENT & MEDICATIONS**

Hour-1 bundle: lactate, cultures, broad-spectrum antibiotics, 30 mL/kg crystalloid for hypotension or lactate 4+, vasopressors for MAP 65+.

**NURSING CARE**

Early recognition, frequent vital signs, MAP and urine output goals, repeat lactate, strict I&O.

**COMPLICATIONS**

Septic shock, ARDS, kidney failure, DIC, death.

**PATIENT EDUCATION**

Know infection warning signs, complete antibiotics, vaccinations, seek care early.

# Urinary Tract Infection

**WHAT IS IT?**

Bacterial infection of the bladder (cystitis) or kidney (pyelonephritis).

**PATHOPHYSIOLOGY**

Bacteria (usually *E. coli*) ascend the urethra into the bladder and may reach the kidneys.

**CAUSES & RISK FACTORS**

Female anatomy, sexual activity, catheters, urinary retention, diabetes, pregnancy, menopause.

**SIGNS & SYMPTOMS**

Burning, frequency, urgency, suprapubic pain, cloudy urine. Pyelonephritis: fever, flank pain. Older adults: confusion.

**DIAGNOSTICS & LABS**

Urinalysis (nitrites, leukocyte esterase), urine culture (clean catch or sterile catheter sample).

**TREATMENT & MEDICATIONS**

Antibiotics (nitrofurantoin, TMP-SMX, cephalosporins, fluoroquinolones per local resistance), fluids.

**NURSING CARE**

Collect culture before antibiotics, remove unnecessary catheters, encourage fluids, assess for sepsis.

**COMPLICATIONS**

Pyelonephritis, urosepsis, kidney damage.

**PATIENT EDUCATION**

Finish antibiotics, wipe front to back, void after intercourse, drink fluids.

# Iron-Deficiency Anemia

**WHAT IS IT?**

Low hemoglobin from not enough iron to make red blood cells.

**PATHOPHYSIOLOGY**

Iron shortage produces small, pale (microcytic, hypochromic) red cells that carry less oxygen.

**CAUSES & RISK FACTORS**

Blood loss (GI, menstrual), pregnancy, poor intake, malabsorption (celiac, bariatric surgery).

**SIGNS & SYMPTOMS**

Fatigue, pallor, dyspnea on exertion, tachycardia, brittle nails, pica, cold intolerance.

**DIAGNOSTICS & LABS**

CBC (low Hgb, low MCV), low ferritin, low serum iron, high TIBC; look for a bleeding source.

**TREATMENT & MEDICATIONS**

Oral iron (with vitamin C), IV iron if needed, treat the cause, transfusion if severe.

**NURSING CARE**

Monitor Hgb and symptoms, fall precautions, assess for bleeding, give liquid iron through a straw.

**COMPLICATIONS**

Heart strain, heart failure, poor pregnancy outcomes, developmental delay in children.

**PATIENT EDUCATION**

Take iron on an empty stomach if tolerated, dark stools are expected, iron-rich foods, keep away from children.

# Sickle Cell Disease

**WHAT IS IT?**

Inherited disorder producing abnormal hemoglobin S that makes red cells sickle.

**PATHOPHYSIOLOGY**

Sickled cells block small vessels (vaso-occlusion) and break down early (hemolysis), causing pain and organ damage.

**CAUSES & RISK FACTORS**

Autosomal recessive inheritance. Crisis triggers: dehydration, infection, cold, hypoxia, stress.

**SIGNS & SYMPTOMS**

Pain crises, fatigue, jaundice, swelling of hands and feet (children), frequent infections.

**DIAGNOSTICS & LABS**

Newborn screening, hemoglobin electrophoresis, CBC, reticulocyte count.

**TREATMENT & MEDICATIONS**

Hydroxyurea, pain control (opioids), IV fluids, oxygen if hypoxic, transfusions, antibiotics, stem cell transplant.

**NURSING CARE**

Prompt pain management, hydration, warmth, monitor for acute chest syndrome and stroke, infection prevention.

**COMPLICATIONS**

Acute chest syndrome, stroke, splenic sequestration, organ damage, infections.

**PATIENT EDUCATION**

Stay hydrated, avoid cold and high altitude, vaccinations, early care for fever.

# Parkinson's Disease

**WHAT IS IT?**

Progressive movement disorder from loss of dopamine-producing neurons.

**PATHOPHYSIOLOGY**

Neurons in the substantia nigra degenerate, reducing dopamine and upsetting the dopamine-acetylcholine balance.

**CAUSES & RISK FACTORS**

Mostly unknown; age, genetics and environmental exposures may contribute.

**SIGNS & SYMPTOMS**

Resting tremor (pill-rolling), rigidity, bradykinesia, shuffling gait, postural instability, masked face.

**DIAGNOSTICS & LABS**

Clinical diagnosis; response to levodopa supports it. Imaging to rule out other causes.

**TREATMENT & MEDICATIONS**

Carbidopa/levodopa, dopamine agonists, MAO-B inhibitors, physical therapy, deep brain stimulation.

**NURSING CARE**

Fall prevention, give meds on time, swallow assessment, nutrition, monitor for orthostatic hypotension.

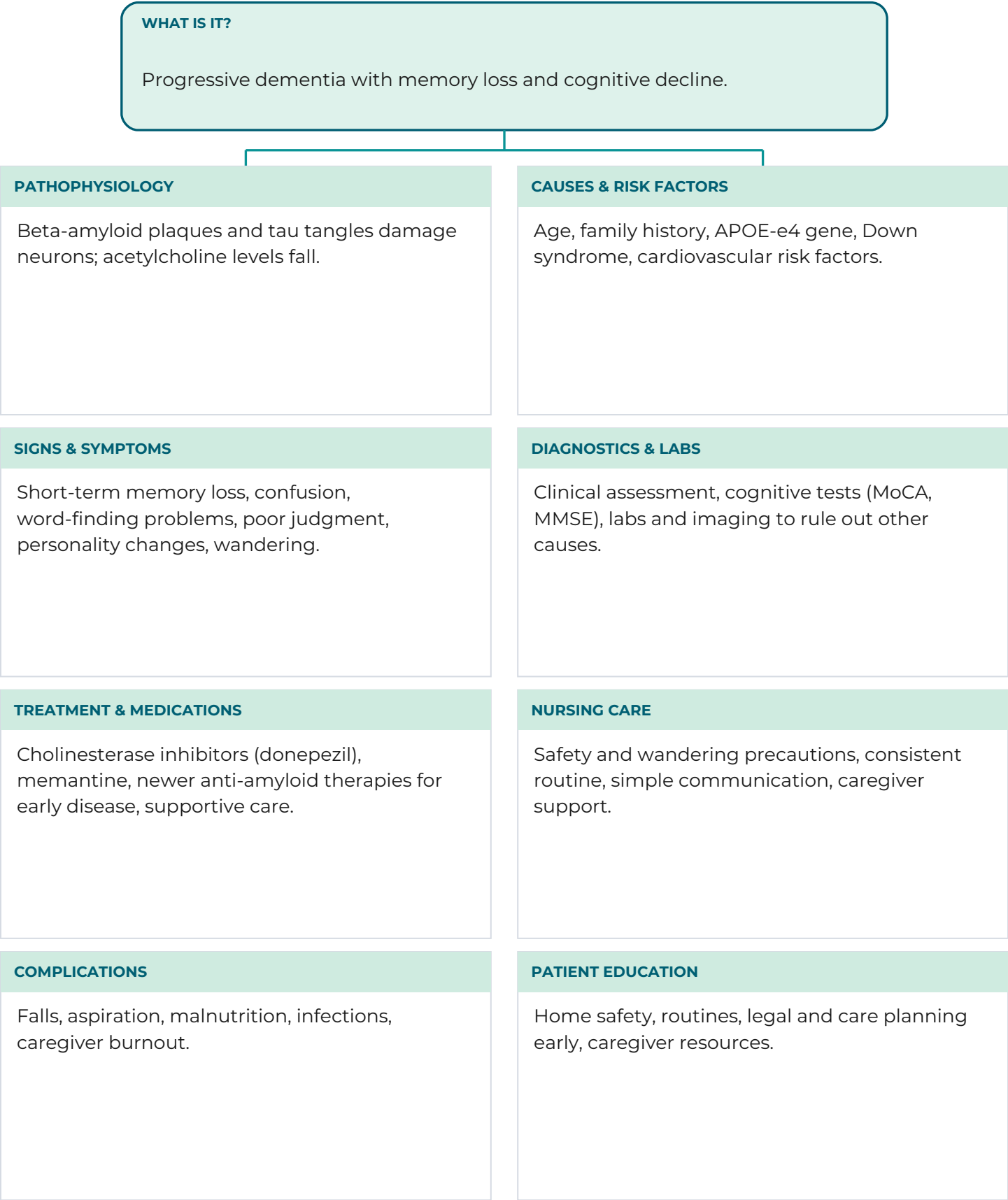
**COMPLICATIONS**

Falls, aspiration pneumonia, dementia, depression.

**PATIENT EDUCATION**

Take meds on schedule, limit high-protein meals near levodopa, home safety, exercise.

# Alzheimer's Disease



# Multiple Sclerosis

**WHAT IS IT?**

Autoimmune demyelination of the brain and spinal cord.

**PATHOPHYSIOLOGY**

Immune attacks destroy myelin, slowing or blocking nerve signals; plaques form scattered through the CNS.

**CAUSES & RISK FACTORS**

Autoimmune; risks include female sex, age 20-40, low vitamin D, smoking, Epstein-Barr virus.

**SIGNS & SYMPTOMS**

Vision problems (optic neuritis), numbness, weakness, fatigue, spasticity, bladder problems, balance issues.

**DIAGNOSTICS & LABS**

MRI (lesions), lumbar puncture (oligoclonal bands), evoked potentials.

**TREATMENT & MEDICATIONS**

Disease-modifying therapies, corticosteroids for relapses, symptom management (baclofen for spasticity).

**NURSING CARE**

Fatigue management, fall prevention, bladder care, skin checks, avoid overheating.

**COMPLICATIONS**

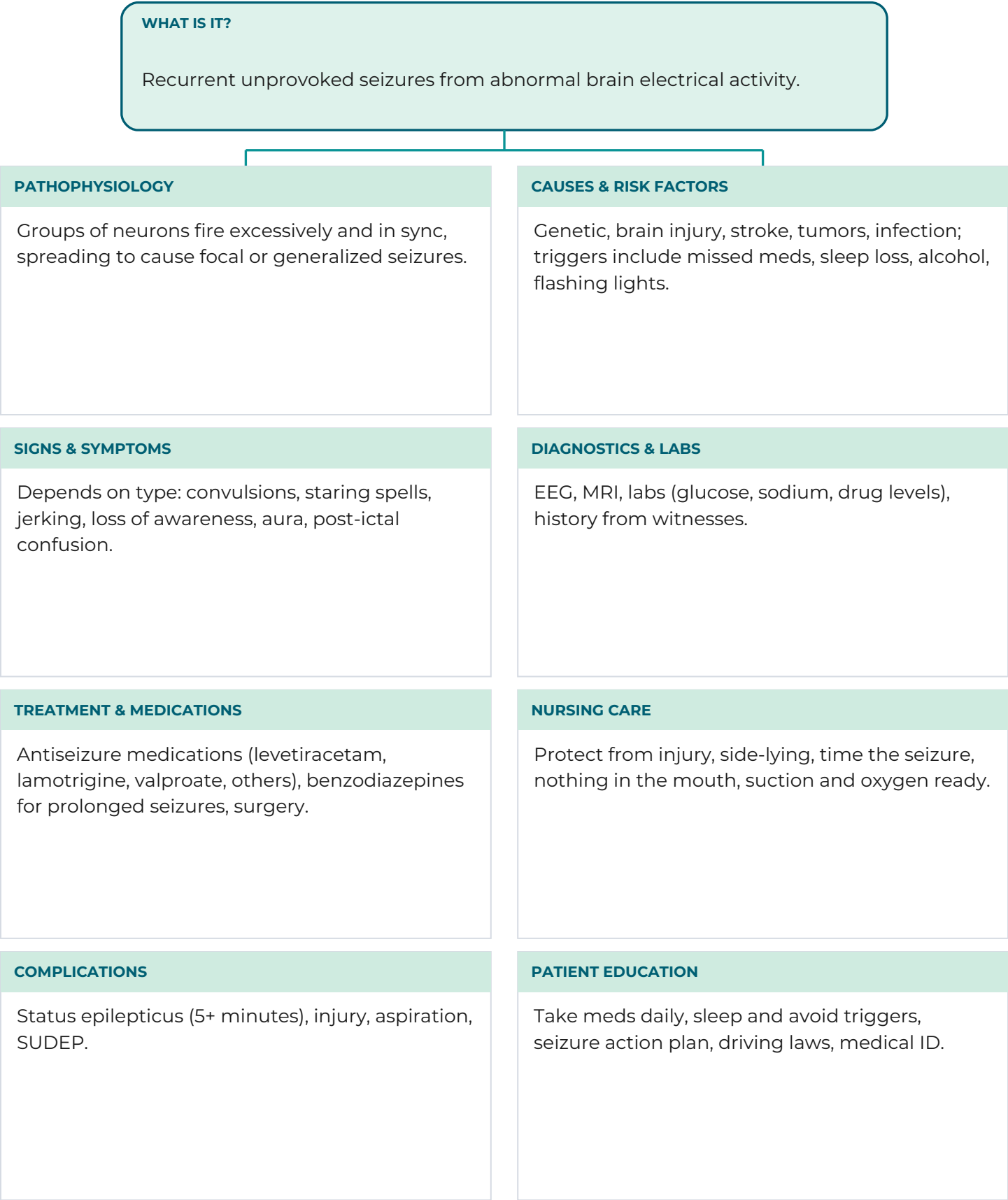
Disability, UTIs, pressure injuries, depression.

**PATIENT EDUCATION**

Avoid heat, pace activities, report relapses, take DMTs as directed.

Seizure Disorder (Epilepsy)

Neurological



# Major Depressive Disorder

**WHAT IS IT?**

Persistent low mood or loss of interest for 2+ weeks, affecting daily function.

**PATHOPHYSIOLOGY**

Involves changes in neurotransmitters (serotonin, norepinephrine, dopamine), stress hormones and brain circuits.

**CAUSES & RISK FACTORS**

Genetics, trauma, chronic stress, medical illness, substance use, certain medications.

**SIGNS & SYMPTOMS**

Depressed mood, anhedonia, sleep and appetite changes, fatigue, worthlessness, poor concentration, suicidal thoughts.

**DIAGNOSTICS & LABS**

Clinical interview (DSM criteria), PHQ-9 screening, rule out thyroid and other causes.

**TREATMENT & MEDICATIONS**

Psychotherapy, SSRIs/SNRIs and other antidepressants, ECT for severe cases, lifestyle support.

**NURSING CARE**

Suicide risk assessment and safety, therapeutic communication, monitor for early antidepressant side effects.

**COMPLICATIONS**

Suicide, substance use, worsening physical health.

**PATIENT EDUCATION**

Antidepressants take weeks to work, do not stop suddenly, crisis line (988 in the US).

# Schizophrenia

**WHAT IS IT?**

Chronic psychotic disorder affecting thinking, perception and behavior.

**PATHOPHYSIOLOGY**

Linked to dopamine excess in some brain pathways and changes in glutamate and brain structure.

**CAUSES & RISK FACTORS**

Genetics, prenatal and birth complications, cannabis use in adolescence, stress.

**SIGNS & SYMPTOMS**

Positive: hallucinations, delusions, disorganized speech. Negative: flat affect, withdrawal, poor motivation.

**DIAGNOSTICS & LABS**

Clinical diagnosis (DSM criteria); rule out substances and medical causes.

**TREATMENT & MEDICATIONS**

Antipsychotics (atypical preferred), long-acting injections, psychosocial therapy, clozapine for resistant cases.

**NURSING CARE**

Safety, reality orientation, do not argue with delusions, monitor for EPS, NMS and metabolic effects.

**COMPLICATIONS**

Suicide, substance use, metabolic syndrome from meds, social isolation.

**PATIENT EDUCATION**

Take medication consistently, report side effects, avoid drugs and alcohol, relapse warning signs.

# Preeclampsia

**WHAT IS IT?**

New high blood pressure after 20 weeks of pregnancy with protein in urine or organ dysfunction.

**PATHOPHYSIOLOGY**

Abnormal placental development causes widespread blood vessel dysfunction and vasospasm.

**CAUSES & RISK FACTORS**

First pregnancy, prior preeclampsia, chronic hypertension, diabetes, multiple gestation, obesity, age over 35.

**SIGNS & SYMPTOMS**

BP 140/90 or higher, headache, visual changes, right upper quadrant pain, edema, hyperreflexia.

**DIAGNOSTICS & LABS**

BP readings, urine protein, platelets, LFTs, creatinine.

**TREATMENT & MEDICATIONS**

Delivery is the cure. Antihypertensives (labetalol, hydralazine, nifedipine), magnesium sulfate for seizure prevention.

**NURSING CARE**

Monitor BP, reflexes, RR and urine output on magnesium; calcium gluconate at bedside; fetal monitoring.

**COMPLICATIONS**

Eclampsia (seizures), HELLP syndrome, placental abruption, stroke.

**PATIENT EDUCATION**

Report headache, vision changes or abdominal pain; low-dose aspirin may be advised in future pregnancies.

# Osteoporosis

**WHAT IS IT?**

Low bone mass and weakened bone structure, raising fracture risk.

**PATHOPHYSIOLOGY**

Bone breakdown outpaces bone formation, especially after menopause, making bones porous and fragile.

**CAUSES & RISK FACTORS**

Age, menopause, low calcium/vitamin D, steroids, smoking, alcohol, inactivity, small frame.

**SIGNS & SYMPTOMS**

Often none until a fracture; height loss, kyphosis, back pain.

**DIAGNOSTICS & LABS**

DEXA scan (T-score -2.5 or lower), fracture history, vitamin D level.

**TREATMENT & MEDICATIONS**

Calcium and vitamin D, weight-bearing exercise, bisphosphonates, denosumab, anabolic agents.

**NURSING CARE**

Fall prevention, safe mobility, teach bisphosphonate administration.

**COMPLICATIONS**

Hip, spine and wrist fractures, chronic pain, loss of independence.

**PATIENT EDUCATION**

Weight-bearing exercise, calcium-rich diet, home safety, take bisphosphonates upright with water.

# Rheumatoid Arthritis

**WHAT IS IT?**

Chronic autoimmune inflammation of the joints.

**PATHOPHYSIOLOGY**

The immune system attacks the synovium, which thickens and erodes cartilage and bone.

**CAUSES & RISK FACTORS**

Autoimmune; risks include female sex, genetics, smoking.

**SIGNS & SYMPTOMS**

Symmetric joint pain and swelling (hands and wrists), morning stiffness over 1 hour, fatigue, nodules.

**DIAGNOSTICS & LABS**

Rheumatoid factor, anti-CCP antibodies, ESR/CRP, X-rays or ultrasound.

**TREATMENT & MEDICATIONS**

DMARDs (methotrexate first-line), biologics, short-term steroids, NSAIDs for symptoms.

**NURSING CARE**

Joint protection, pain management, monitor for infection and methotrexate toxicity.

**COMPLICATIONS**

Joint deformity, disability, heart disease, lung disease, infections.

**PATIENT EDUCATION**

Balance rest and activity, heat for stiffness, take methotrexate weekly not daily, avoid live vaccines on biologics.

# Cushing's Syndrome

**WHAT IS IT?**

Excess cortisol in the body.

**PATHOPHYSIOLOGY**

Too much cortisol raises glucose, breaks down protein, redistributes fat and suppresses immunity.

**CAUSES & RISK FACTORS**

Long-term corticosteroid use (most common), pituitary adenoma (Cushing's disease), adrenal tumors.

**SIGNS & SYMPTOMS**

Moon face, buffalo hump, central obesity, purple striae, thin skin, hyperglycemia, hypertension, weakness.

**DIAGNOSTICS & LABS**

24-hour urine cortisol, late-night salivary cortisol, dexamethasone suppression test, ACTH level, imaging.

**TREATMENT & MEDICATIONS**

Taper steroids if the cause, surgery for tumors, medications that lower cortisol.

**NURSING CARE**

Monitor glucose, BP, fluid balance, infection signs and skin integrity; fall precautions.

**COMPLICATIONS**

Diabetes, infections, osteoporosis, fractures, heart disease.

**PATIENT EDUCATION**

Never stop steroids abruptly, monitor glucose, infection prevention, medical ID.

# EDUCATIONAL USE ONLY

## NOT A CLINICAL REFERENCE

This study aid is a learning and exam-review tool for nursing and pharmacy students. It must not be used to make patient-care decisions, including dosing, administration, or monitoring.

## ALWAYS VERIFY

Drug information, boxed (black box) warnings, and high-alert designations change over time. Before applying anything in practice, check the current FDA-approved labeling, your facility's policies, and an up-to-date drug reference. Normal lab ranges vary between laboratories; use the ranges your program or facility provides.

## RECOMMENDED REFERENCES FOR VERIFICATION

- FDA drug labels: DailyMed ([dailymed.nlm.nih.gov](https://dailymed.nlm.nih.gov))
- ISMP List of High-Alert Medications ([ismp.org](https://www.ismp.org))
- A current nursing drug handbook or your program's required drug reference
- Your course textbook and instructor guidance

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